Hint: Show first that $\mathrm{Tor}_i^A(M,N)$ can be computed by tensoring two projective resolutions, for M and for N, and computing cohomology.

(v) Let $0 \to M_1 \to M_2 \to M_3 \to 0$ be a short exact sequence of left A-modules. Let $P^1_\bullet$ and $P^3_\bullet$ be projective resolutions of M_1 and M_3. Construct a projective resolution of M_2 with terms $P^2_i := P^1_i \oplus P^3_i$ (use Exercise 8.1.4). Use it to show that for any left A-module N there are long exact sequences

$$0 \to \mathrm{Hom}_A(M_3,N) \to \mathrm{Hom}_A(M_2,N) \to \mathrm{Hom}_A(M_1,N) \to$$
$$\mathrm{Ext}^1(M_3,N) \to \mathrm{Ext}^1(M_2,N) \to \mathrm{Ext}^1(M_1,N) \to \mathrm{Ext}^2(M_3,N) \to \dots$$

and

$$\dots \to \mathrm{Tor}_2(M_3,N) \to \mathrm{Tor}_1(M_1,N) \to \mathrm{Tor}_1(M_2,N) \to \mathrm{Tor}_1(M_3,N)$$
$$\to M_1 \otimes_A N \to M_2 \otimes_A N \to M_3 \otimes_A N \to 0.$$

Problem 8.2.7. (i) Let $A = \mathbb{Z}$ and let M, N be finitely generated abelian groups. Compute $\mathrm{Tor}_i(M,N)$, $\mathrm{Ext}^i(M,N)$ (Hint: Reduce to the case of cyclic groups using the classification theorem for finite abelian groups.)

(ii) Do the same for $A = k[x]$ and M, N being any finitely generated A-modules.

Problem 8.2.8. Show that if A_1, A_2 are algebras over a field k and M_i, N_i are left A_i-modules, then

$$\mathrm{Tor}_i^{A_1\otimes A_2}(M_1\otimes M_2, N_1\otimes N_2) =$$
$$\bigoplus_{j+m=i} \mathrm{Tor}_j^{A_1}(M_1,N_1)\otimes \mathrm{Tor}_m^{A_2}(M_2,N_2).$$

Similarly,

$$\mathrm{Ext}^i_{A_1\otimes A_2}(M_1\otimes M_2, N_1\otimes N_2) =$$
$$\bigoplus_{j+m=i} \mathrm{Ext}^j_{A_1}(M_1,N_1)\otimes \mathrm{Ext}^m_{A_2}(M_2,N_2),$$

if N_i are finite dimensional.

In a similar way one can define Tor and Ext groups in any abelian category which has **enough projectives**; i.e., any object is a quotient of a projective object (this condition insures that every object has a projective resolution).